

discovery

Multi-Mouse Habitat Discovery

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Date : 2026-08-04 14:48
Mice analysed : 10
Method : GMM
K selection : silhouette elbow criterion
Chosen K : 4
Parameters : DTI-AD, DTI-RD, DTI-FA, T2, T2star, MTR
2nd pass : suggest (cosine $\geq$ 0.92)
→ Merge candidate: MH3 + MH4

Discovery is purely data-driven – no biological labels are used during meta-clustering.
User labels appear ONLY in F8 (validation).

Silhouette scores by K:

K=2: 0.467 K=3: 0.538 K=4: 0.506 K=5: 0.509 K=6: 0.442
(max silhouette at K=3: 0.538 – not chosen;
elbow avoids K=2 outlier trap from extreme clusters)

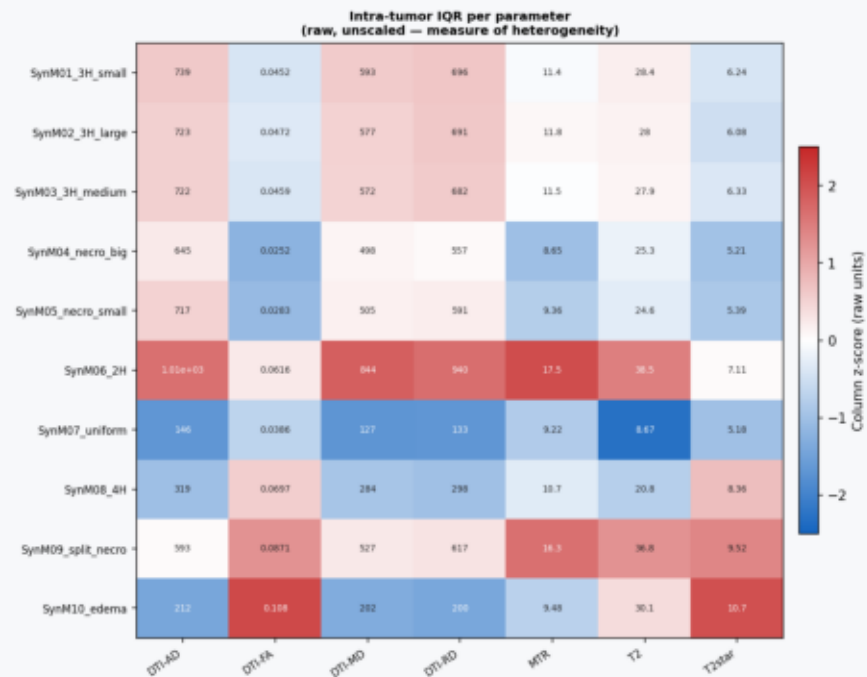
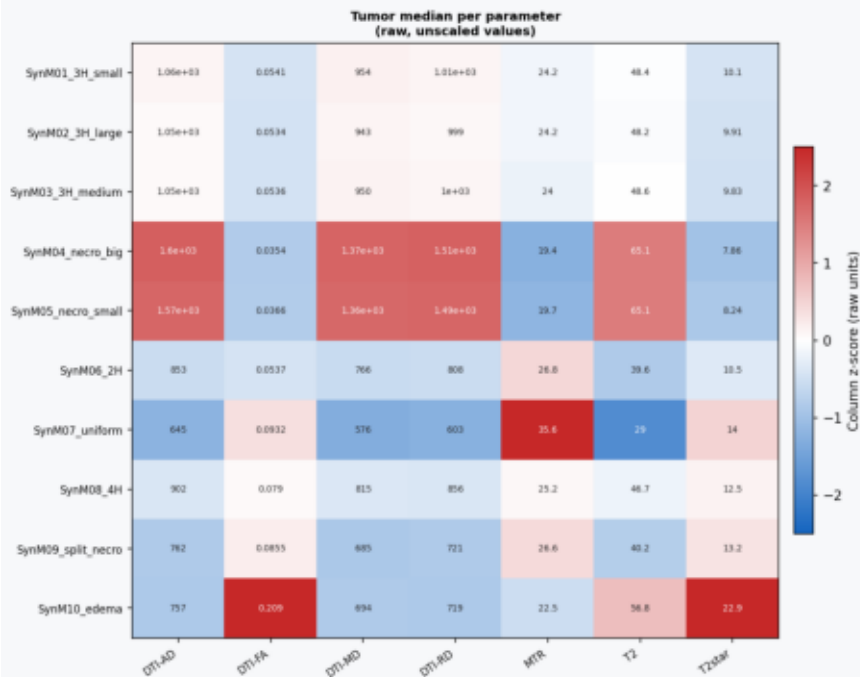
Meta-habitats (sorted by prevalence):

MH1	10/10 mice	19 clusters	presence	100.0%
MH2	6/10 mice	8 clusters	presence	60.0%
MH3	5/10 mice	6 clusters	presence	50.0%
MH4	2/10 mice	2 clusters	presence	20.0%

Figures:

F0.0 – Cohort tumor profile (raw medians & IQR)
F1 – Meta-habitat summary table
F2 – K selection curve
F3 – PCA scatter (centroid space)
F4 – Mean centroid heatmap
F5 – Per-MH MRI bar profiles
F6 – Radar profiles (per meta-habitat)
F7 – Radar profiles (all overlaid)
F8 – User label vs MH cross-tabulation
F_{cos} – Directional similarity (2nd pass)
F9 – Segmentation maps

Cohort tumor profile — absolute position in parameter space
 Colour = column z-score across mice · Cell = raw value
 Blue = low signal · Red = elevated signal



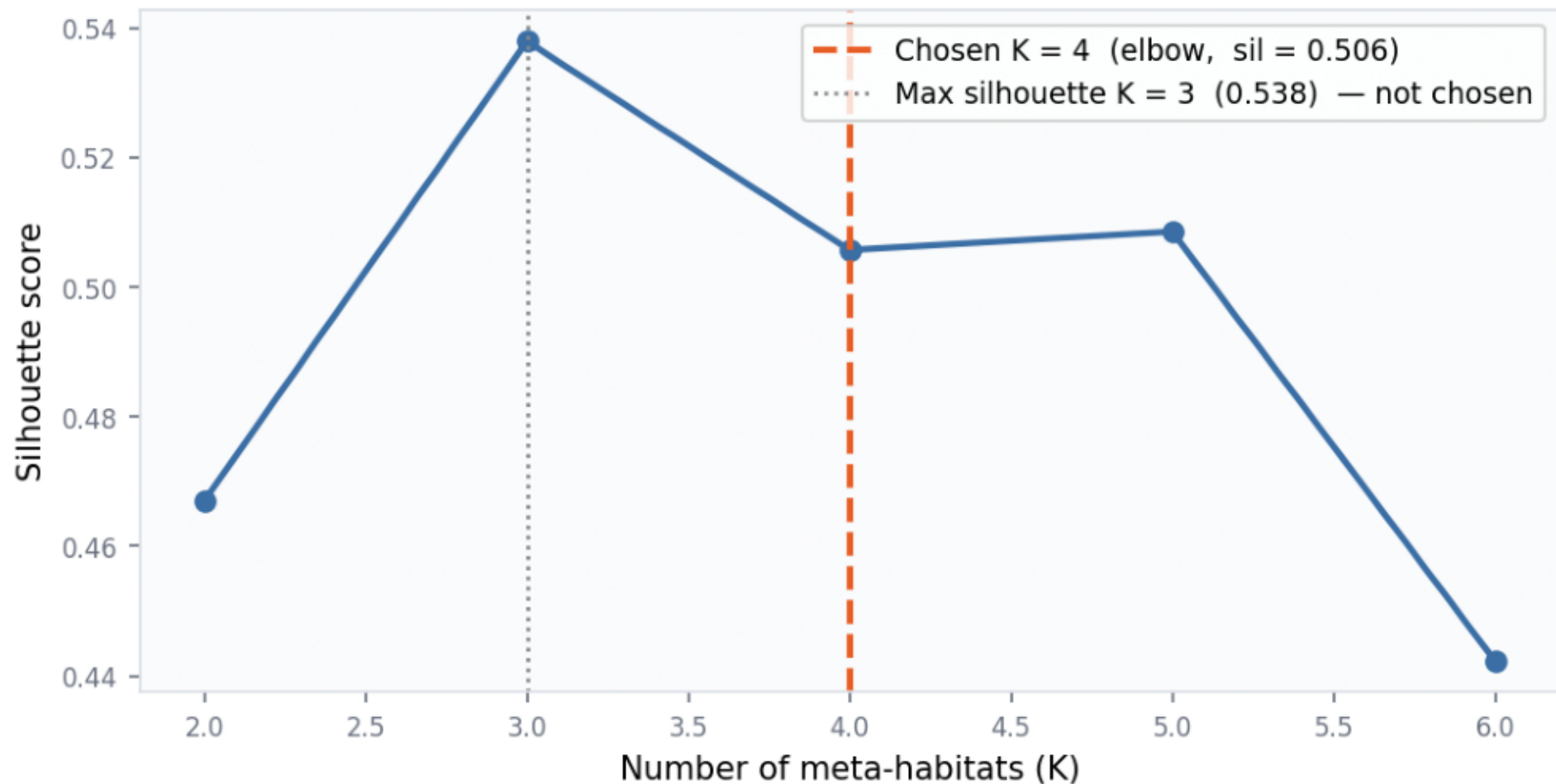
**Mouse index — M# labels are used throughout this report
(cluster compositions, radar chart traces)**

#	Mouse name	Full path (.../parent/folder)	K
M1	SynM01_3H_small	../_discovery_test/classic/SynM01_3H_small	5
M2	SynM02_3H_large	../_discovery_test/classic/SynM02_3H_large	5
M3	SynM03_3H_medium	../_discovery_test/classic/SynM03_3H_medium	6
M4	SynM04_necro_big	../_discovery_test/classic/SynM04_necro_big	2
M5	SynM05_necro_small	../_discovery_test/classic/SynM05_necro_small	2
M6	SynM06_2H	../_discovery_test/classic/SynM06_2H	2
M7	SynM07_uniform	../_discovery_test/classic/SynM07_uniform	2
M8	SynM08_4H	../_discovery_test/classic/SynM08_4H	6
M9	SynM09_split_necro	../_discovery_test/classic/SynM09_split_necro	3
M10	SynM10_edema	../_discovery_test/classic/SynM10_edema	2

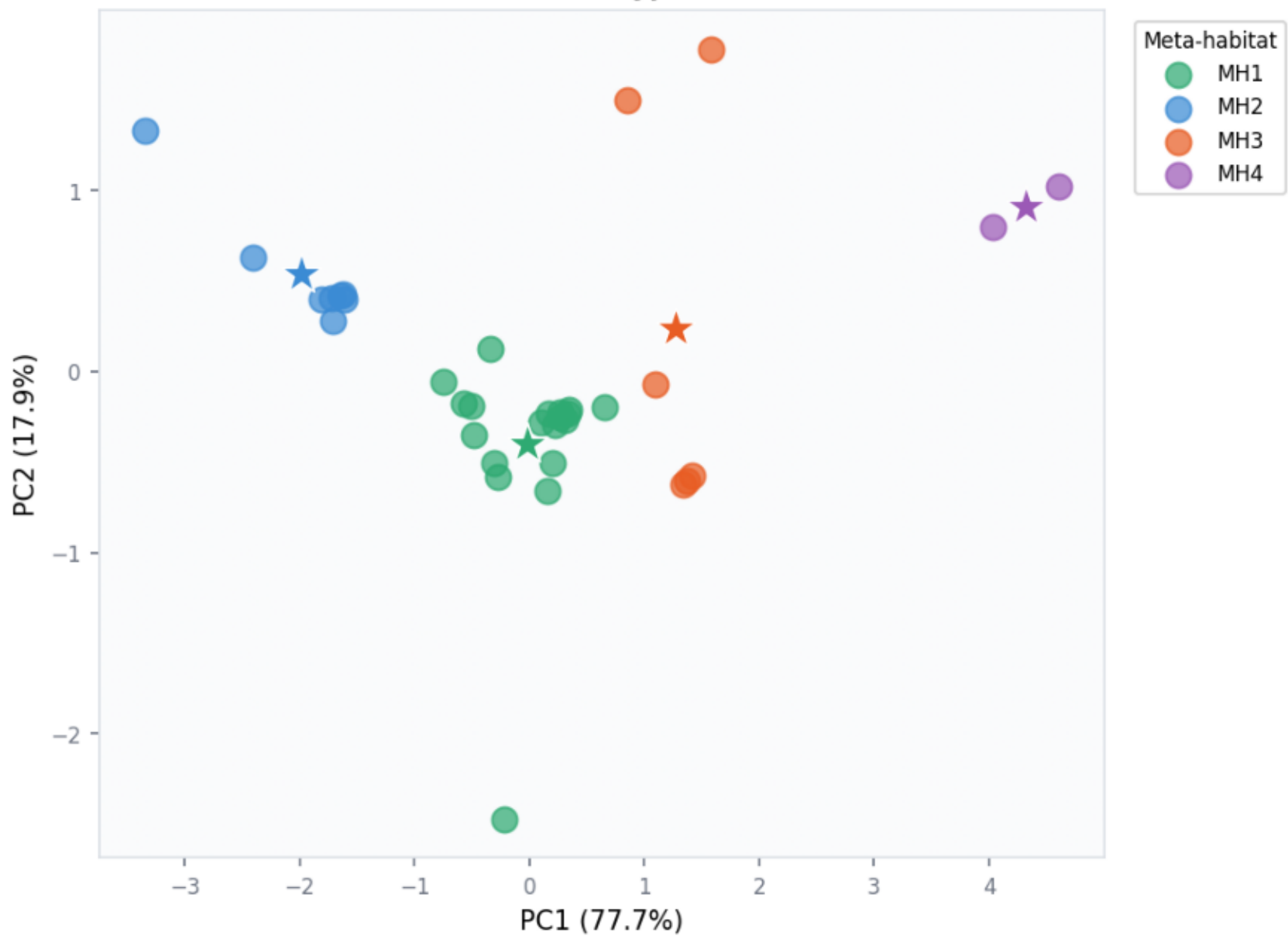
Meta-habitat summary (MRI values in IQR units — 0 = tumour median)
Red = elevated · Blue = decreased · White = near tumour median

Meta-hab.	Mice	Clusters	Prev. %	DTI-AD	DTI-RD	DTI-FA	T2	T2star	MTR
MH1	10/10	19	100%	-0.10 ±0.18	-0.10 ±0.17	+0.11 ±0.37	-0.18 ±0.23	+0.06 ±0.27	+0.25 ±0.37
MH2	6/10	8	60%	+1.09 ±0.55	+1.08 ±0.54	-0.47 ±0.10	+0.74 ±0.20	-0.44 ±0.20	-0.46 ±0.09
MH3	5/10	6	50%	-0.43 ±0.22	-0.44 ±0.23	+1.19 ±0.52	-0.12 ±0.62	+0.90 ±0.31	+0.46 ±0.61
MH4	2/10	2	20%	-1.26 ±0.07	-1.41 ±0.03	+3.71 ±0.28	-0.78 ±0.01	+1.97 ±0.11	+1.13 ±0.11

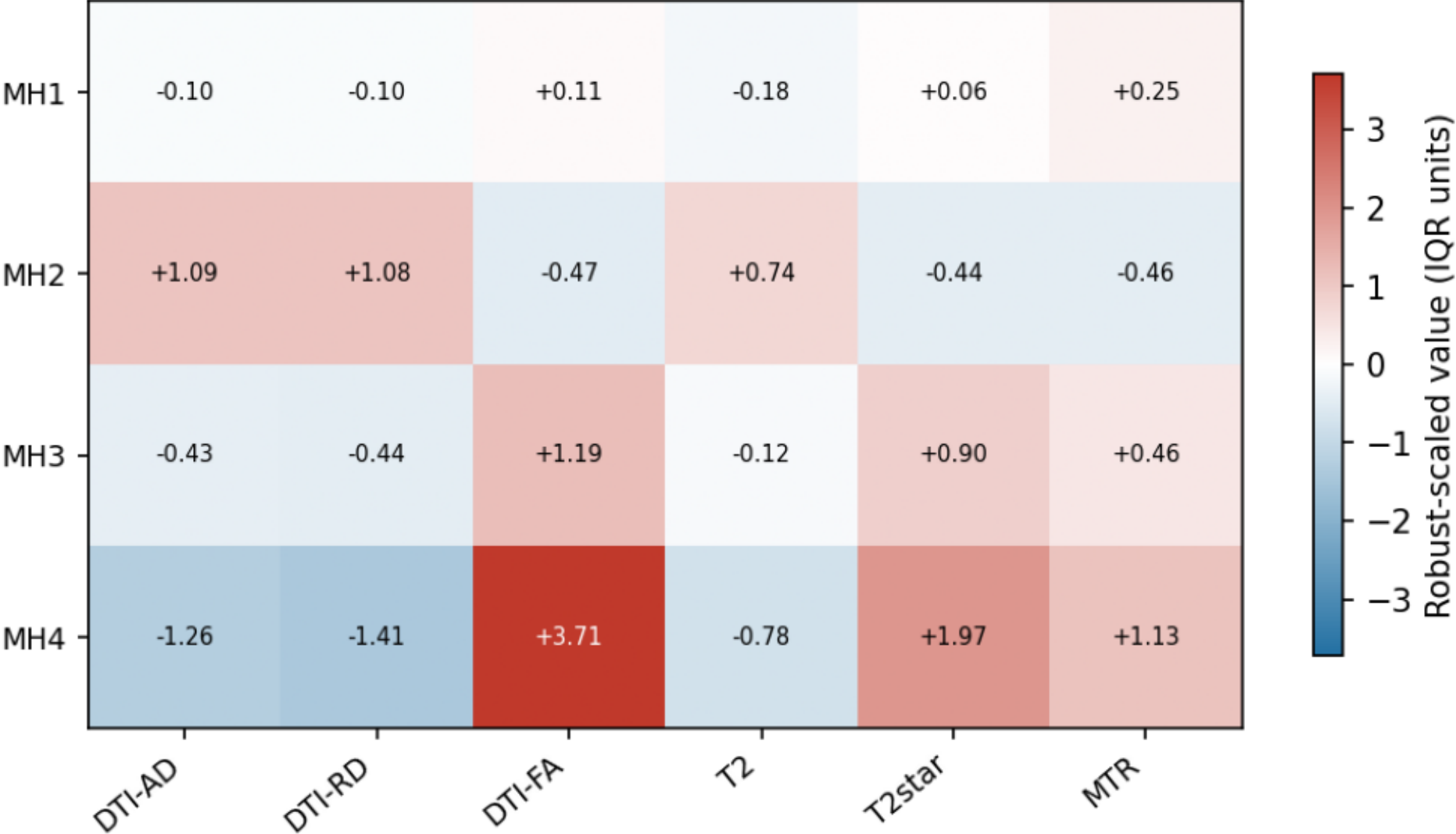
Meta-clustering — K selection (elbow criterion: K where steep descent gives way to flat plateau)



PCA of cluster centroids — meta-habitat coloring
★ = meta-habitat archetype (mean centroid)

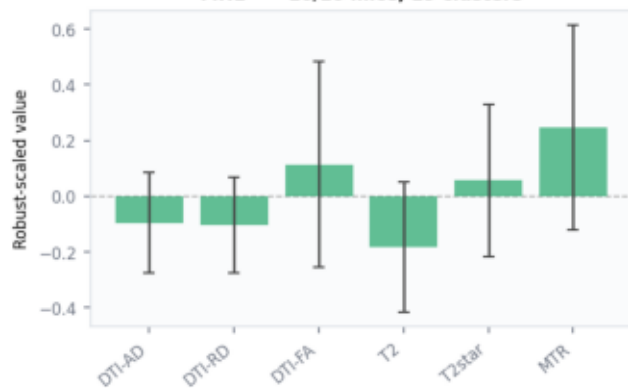


Mean centroid per meta-habitat (robust-scaled, 0 = tumour median)

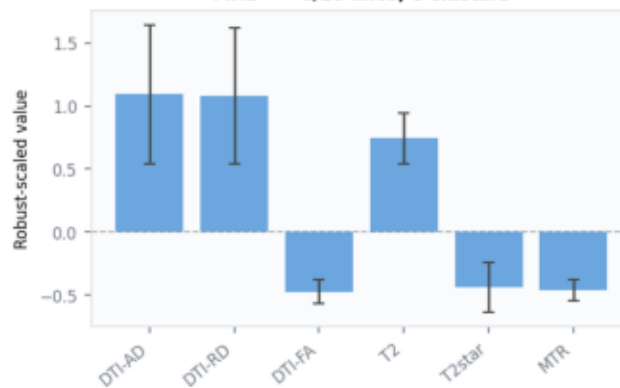


Per-meta-habitat MRI profiles (mean $\pm$ std, robust-scaled)

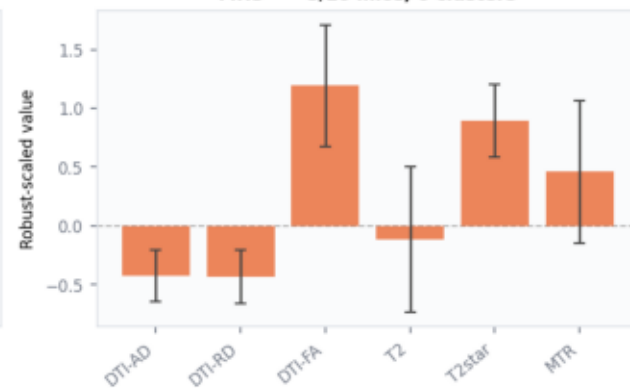
MH1 — 10/10 mice, 19 clusters



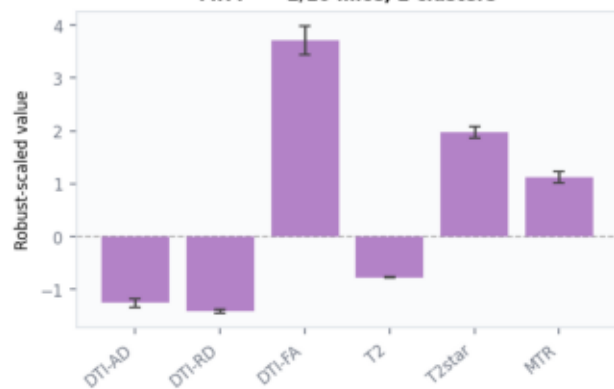
MH2 — 6/10 mice, 8 clusters



MH3 — 5/10 mice, 6 clusters

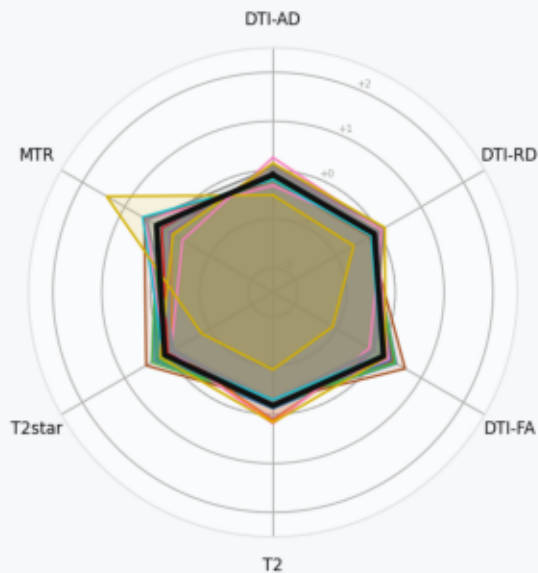


MH4 — 2/10 mice, 2 clusters

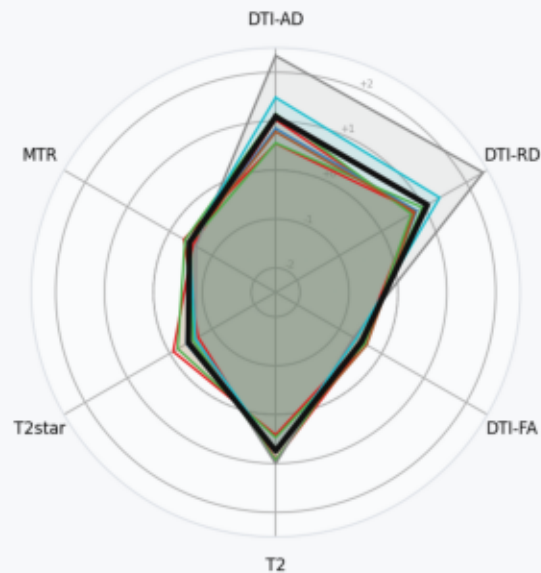


Individual cluster profiles per meta-habitat
(IQR units — dashed ring = tumour median [0] — outliers visible as traces far from group)

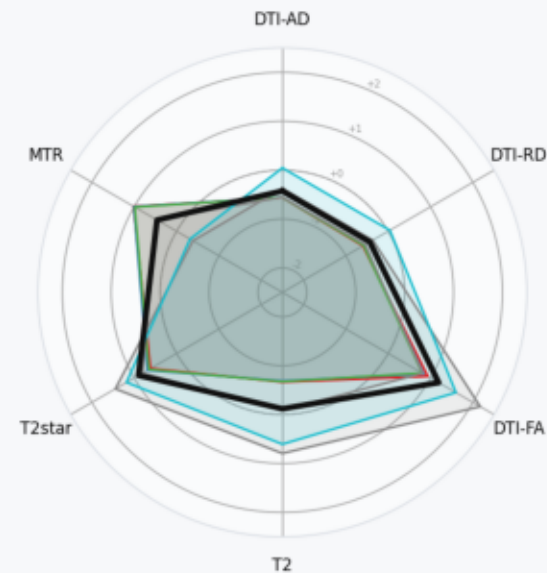
MH1 · 10/10 mice · 19 clusters



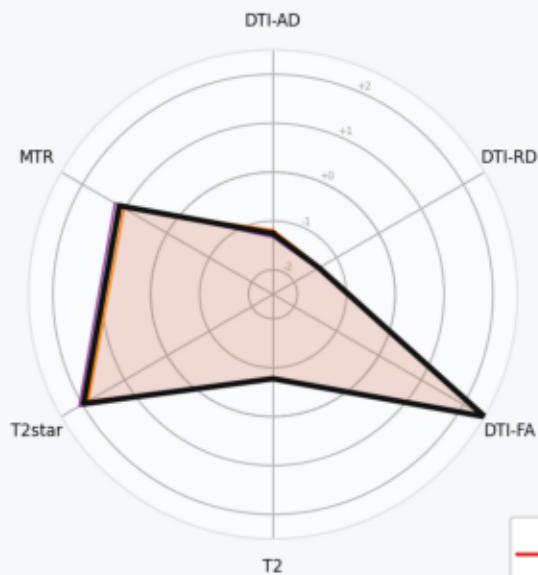
MH2 · 6/10 mice · 8 clusters



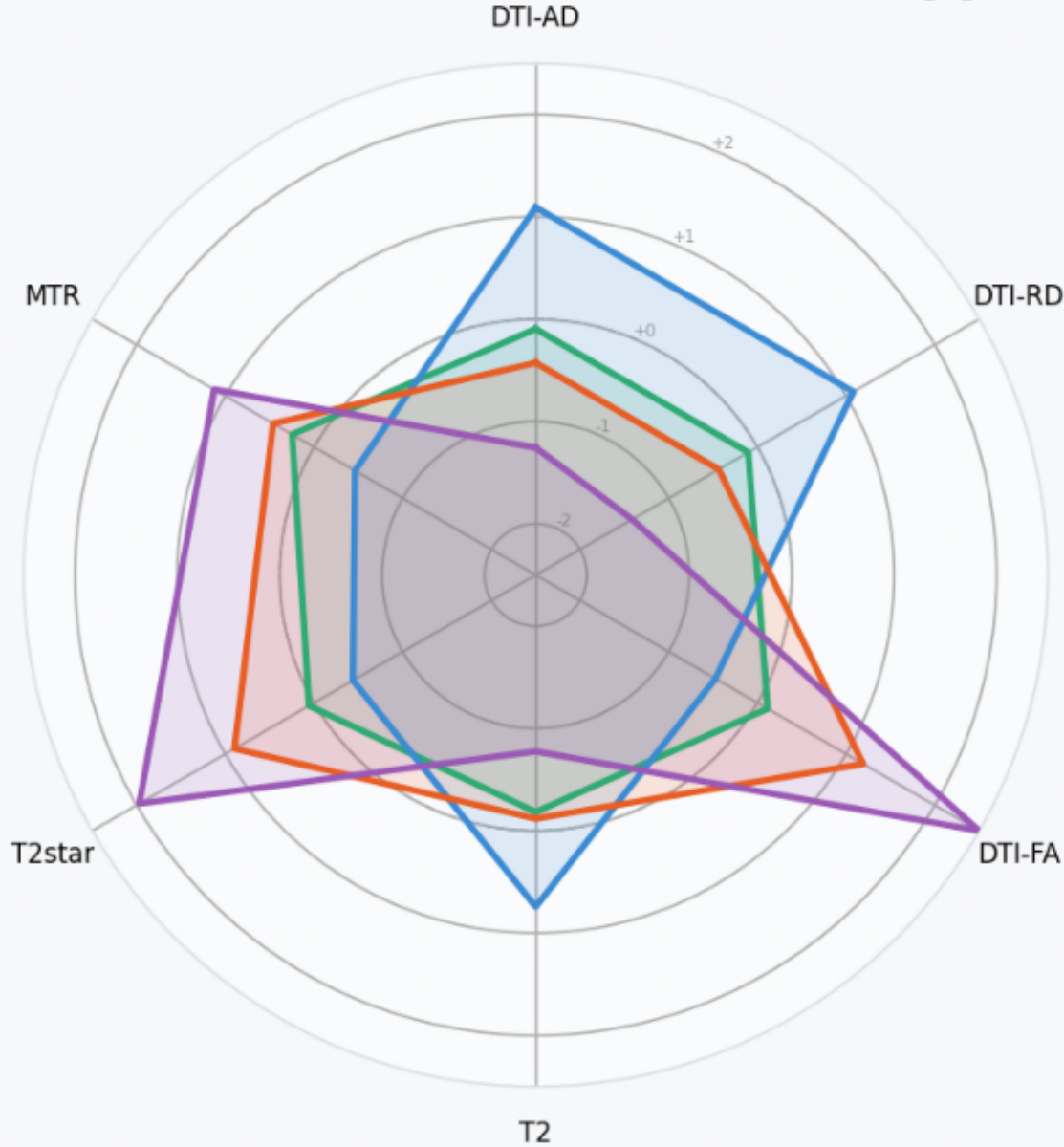
MH3 · 5/10 mice · 6 clusters



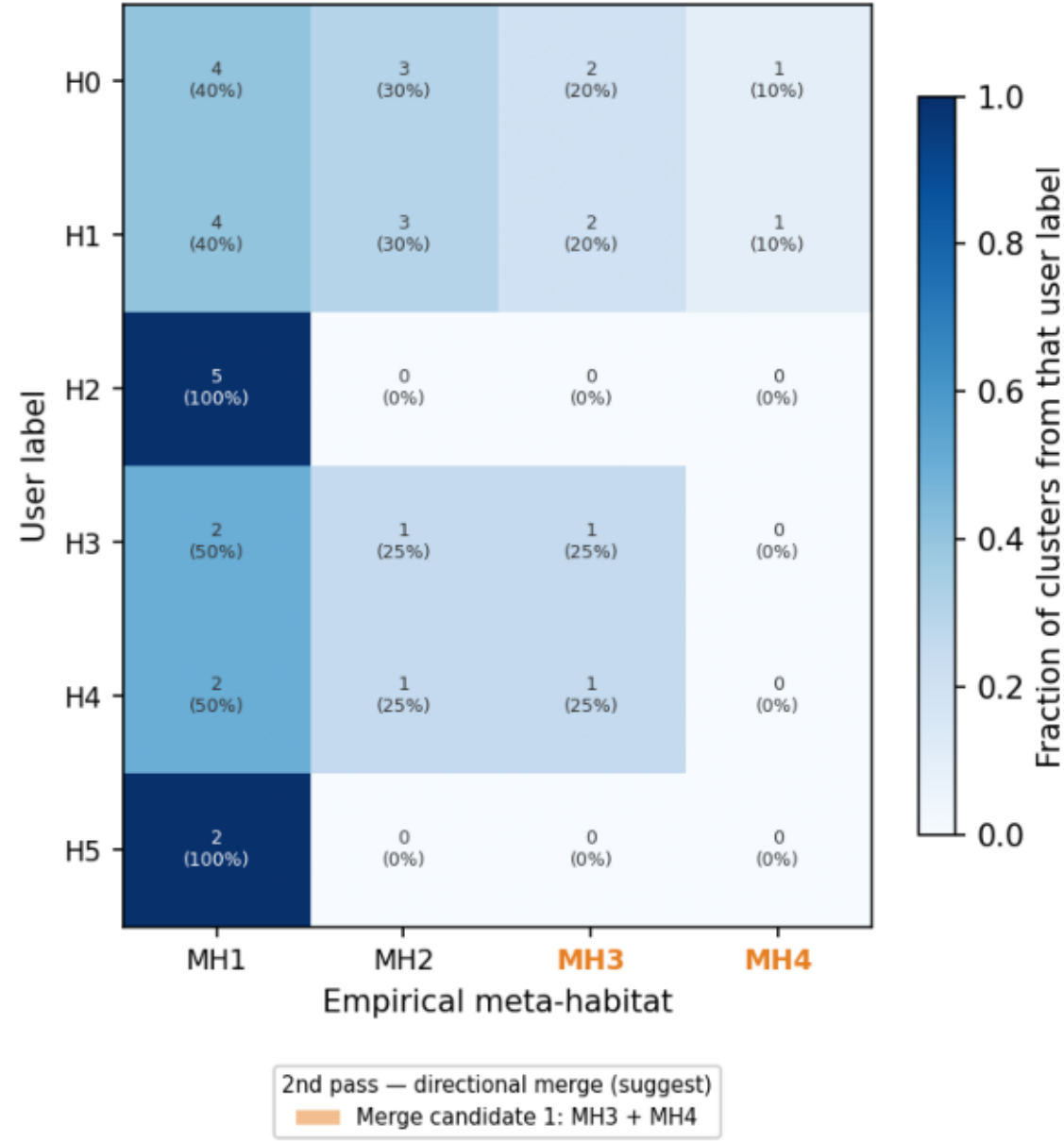
MH4 · 2/10 mice · 2 clusters



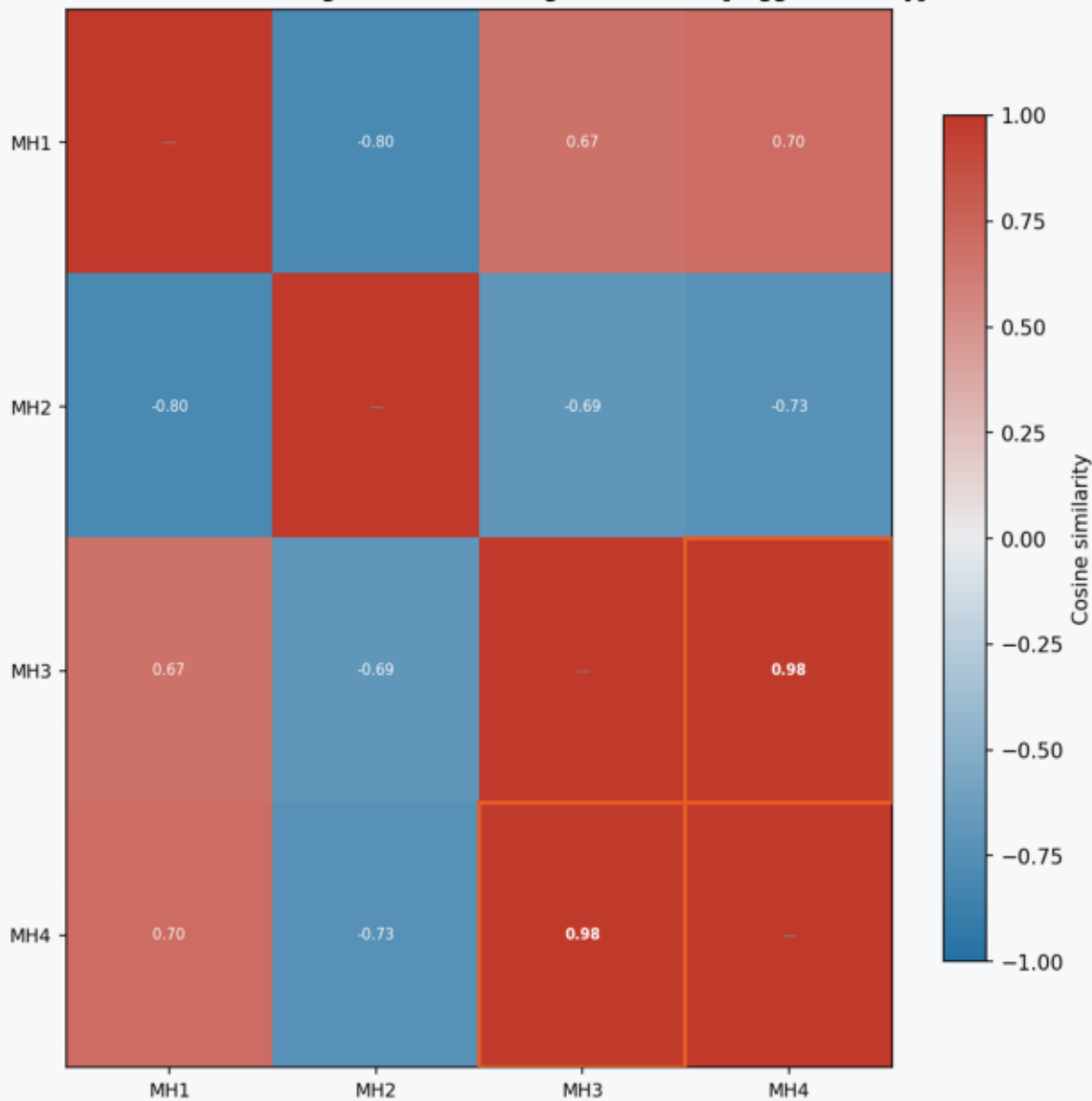
**Combined radar — mean MRI profiles per meta-habitat
(IQR units — dashed circle = tumour median [0])**



Validation: user label → meta-habitat assignment
(row-normalised; labels were NOT used during discovery)

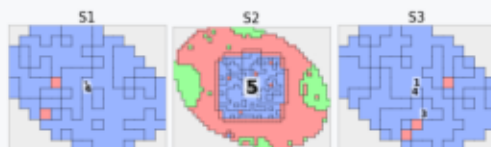


Directional similarity between meta-habitat centroids (cosine)
Threshold = 0.92 · orange-bordered = merge candidates [suggestion only]

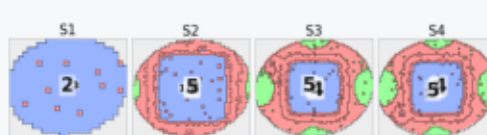


Merge candidates (same direction, different amplitude):
 MH3 + MH4 [DTI-FA_r T2star_r MTR_r / DTI-RD_r DTI-AD_r T2_r] (MH3-MH4: 0.98)

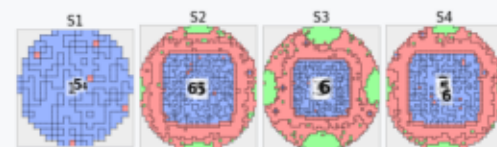
M1



M2



M3



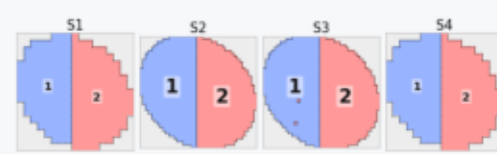
M4



M5

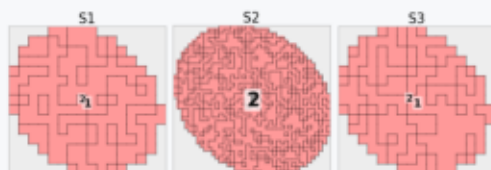


M6

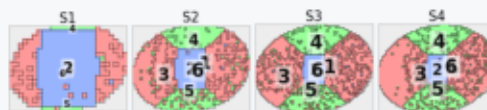


Colour = meta-habitat · numbers = cluster id (1-based)
 MH3 MH2 MH1 MH4

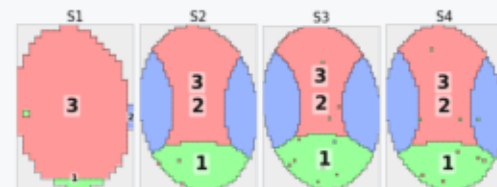
M7



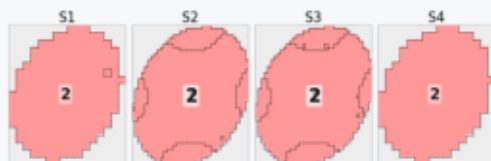
M8



M9



M10



Colour = meta-habitat · numbers = cluster id (1-based)
 MH3 MH2 MH1